Evolutionary Weightless Neural Networks for Network Intrusion Detection

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Abstract. Weightless Neural Networks (WNNs) based on Random Access Memory (RAM) neurons offer constant-time inference with sub-microsecond lookup, making them attractive for edge-deployed Intrusion Detection Systems (IDS). However, their classification performance depends on *connectivity*, which input bits each neuron observes, a problem traditionally solved by random initialization. We present an evolutionary optimization framework that optimizes neuron count through grid search and genetic algorithm, with each neuron's random connectivity effectively performing feature selection intrinsic to the architecture. All our configurations fit within the logic fabric of a \$20 Zynq Z-7020 Field-Programmable Gate Array (FPGA), with our dense networks enabling $O(1)$ direct-lookup inference per neuron. We evaluate on three standard IDS benchmarks (UNSW-NB15, CICIDS2017, CIC-IoT-2023) using temporal and/or random splits with a seven-mode threshold calibration protocol that exposes the sensitivity of binary classifiers to decision boundary selection. Across 30 to 112 independent runs per configuration, our approach achieves 88.87% F1-Macro (F1) at 8.77% false-positive rate (FPR) on UNSW-NB15 temporal (16-bit thermometer, validation-calibrated threshold), outperforming Random Forest by 2.82 points F1 at roughly one-third the FPR, and 99.55% F1 on CICIDS2017 random (0.12% FPR). On the full 46.7M-record CIC-IoT-2023 benchmark, a 25 KB dense configuration achieves 83.13% F1 at 2.06% FPR in just 304 logic elements, while sparse configurations reach 84.79% F1 at 1.59% FPR. All designs are verified through Vivado synthesis on the Zynq Z-7020.

Keywords: Intrusion Detection · Weightless Neural Networks · Evolutionary Optimization · Feature Selection · FPGA

1 Introduction

Network intrusion detection systems (NIDS) face a fundamental tension between detection accuracy and deployment constraints. Deep learning models achieve high accuracy but require megabytes of parameters, GPU hardware, and inference times orders of magnitude above those of lookup-based models [7] — far beyond the ∼80 ns per-packet budget at 100 Gbps line rate, making them impractical for inline packet classification on edge devices. Classical ML baselines such as Random Forest [5] and XGBoost [6] run in microseconds but typically require megabytes of model state. Conversely, lightweight signature/rule-based systems such as Snort [19] and Suricata [15] operate at wire speed but lack the adaptability to detect novel attack patterns.

Weightless Neural Networks (WNNs) [1,12] offer a compelling middle ground: RAM-based neurons perform classification via direct memory lookup in $O(1)$ time with model sizes measured in bytes rather than megabytes. Recent work [21] demonstrated WNN-based IDS achieving 98.5% accuracy on UNSW-NB15 with a model of just 192 bytes. Additionally, neuron connectivity (which input bits each neuron observes) plays a critical role in WNN architectures but is traditionally generated randomly. Garcia and de Souto [10] showed that global optimization of connectivity outperforms random initialization, but their evaluation was focused on standard classification benchmarks.

Our work improves on two fronts:

- *How the network is derived.* We introduce an optimization framework using genetic algorithm to discover optimal neuron counts. This optimization effectively performs automated feature selection intrinsic to the WNN architecture.
- *How the network is evaluated.* We test the resulting networks against unseen data using temporal or random splits across three datasets with a threshold calibration study that quantifies the gap between train-time and deployment-time performance.

Our contributions are:

1. **Evolved connectivity as feature selection.** Optimizing network parameters discovers which input features matter without external feature selection with connectivity optimization providing the largest single improvement in classification performance.
2. **Compact architecture with sub-microsecond inference.** Our evolved WNN outperforms Random Forest and XGBoost on the UNSW-NB15 temporal split and achieves 5.9× lower FPR than RF on CIC-IoT-2023, in a 25 KB model deployable on a $20 FPGA.
3. **Seven-mode threshold calibration study.** We identify threshold selection as an important factor in IDS evaluation, comparing seven calibration methods across all datasets and splits.
4. **Three-dataset evaluation with FPGA synthesis.** Results on UNSW-NB15 (temporal and random), CICIDS2017 (random), and CIC-IoT-2023 (random) demonstrate architecture generalizability, with Vivado-verified FPGA deployment on a Zynq Z-7020 for all datasets.

2 Background

2.1 RAM Neurons and Weightless Neural Networks

Unlike conventional neural networks that learn continuous weights through gradient descent, Weightless Neural Networks (WNNs) [1] use Random Access Memory (RAM) neurons that perform classification via direct memory lookup. A

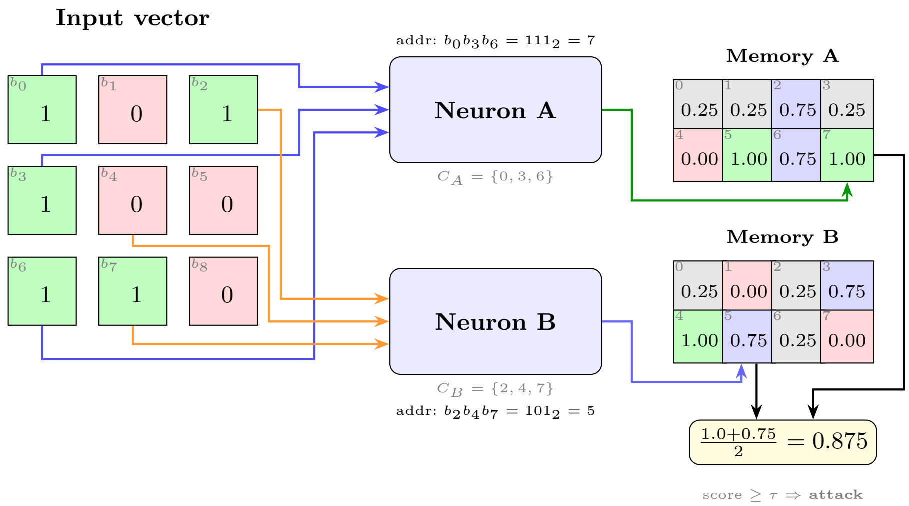

Fig. 1. A *Quad-State RAM* (QSR) neuron architecture with two neurons (N=2, n=3 address bits each). Each neuron selects 3 bits from the 9-bit input via its connectivity map, forms a 3-bit address, and looks up its QSR memory (2^3=8 cells). Cell colors: TRUE (1.0), WEAK_TRUE (0.75), WEAK_FALSE (0.25), FALSE (0.0). The mean score is thresholded to classify the input.

RAM neuron consists of 2^n memory cells, where n is the number of observed input bits, the neuron's *address width*. Given a binary input vector of length d, the neuron selects n bits through a fixed *connectivity map* $C = \{c_1, \ldots, c_n\}$ where $c_i \in \{0, \ldots, d{-}1\}$, concatenates them to form an n-bit address, and returns the stored value at that address. Training writes target labels to addressed cells; inference reads them in $O(1)$ time with no multiply-accumulate operations.

The fundamental mechanism enabling WNN generalization is *partial connectivity* [1,12]. A fully-connected neuron ($n = d$) memorizes every training example without generalizing, since each input maps to a unique address. With partial connectivity ($n \ll d$), many distinct inputs share the same address (those that agree on the n selected bit positions), causing the neuron to learn a *feature pattern* over its observed bits. An ensemble of N neurons with diverse connectivity maps provides N complementary views of the input, and their aggregated scores produce a classification decision. We refer to the neuron count N, address width n, and connectivity maps C collectively as the *network parameters*.

Traditional RAM neurons store binary values (TRUE/FALSE), while Probabilistic Logic Nodes (PLN) [12] add an *undefined* state (u) for untrained cells. We introduce *Quad-State RAM* (QSR) memory (Figure 1) with four states encoded in just 2 bits: FALSE (0.0), WEAK_FALSE (0.25), WEAK_TRUE (0.75), and TRUE (1.0). Unlike PLN's symmetric undefined state (0.5), QSR cells default to WEAK_FALSE (0.25), biasing untrained neurons toward the majority class (normal traffic). When a cell trained as TRUE is subsequently addressed with a normal-class example, it transitions to WEAK_TRUE, preserving partial confidence from both classes. This provides smoother probability estimates than binary voting while requiring only 2 bits per cell, matching PLN's compactness with richer representation.

2.2 Thermometer Encoding

WNNs require binary inputs, but network flow features (packet counts, byte rates, inter-arrival times) are continuous, usually encoded as float64. We convert each feature to a binary vector using *thermometer encoding*, a standard technique in modern WNNs [23]. Given a feature value x and k thresholds $\theta_1 < \cdots < \theta_k$, the encoding produces k bits: bit i is 1 if $x \geq \theta_i$, else 0. This preserves ordinal relationships: similar values differ by few bits, enabling RAM neurons to generalize across nearby feature values. This ordinal encoding also provides natural noise robustness: a small measurement error shifts at most one threshold boundary, flipping a single bit in the encoded vector.

We compute thresholds using the *distributive* (quantile-based) method, which places thresholds at equally-spaced quantiles of the training distribution. This is robust to skewed distributions common in network traffic (e.g., packet sizes follow a heavy-tailed distribution). With k bits per feature and a feature set F, the total input width is $d = k \times |F|$ bits.

2.3 Evaluation Datasets

We evaluate on three standard IDS benchmarks, each with distinct characteristics:

UNSW-NB15 [14] contains network flow records from the Australian Centre for Cyber Security with 42 features (39 numeric, 3 categorical) in the official partition and nine attack categories plus normal traffic. The standard temporal split (175K train / 82K test) separates training and testing periods, preventing data leakage. We also evaluate on a deduplicated random 80/20 split (1.27M / 317K) for comparison with prior work [21]. Both splits are published with top-20 feature rankings.

CICIDS2017 [20] comprises five days of network traffic captured at the Canadian Institute for Cybersecurity with 14 attack types; we use the widely adopted MachineLearningCSV distribution, which retains 78 of the 80+ CICFlowMeter flow features after dropping identifier columns. We use a stratified 80/20 random split (2.3M train / 566K test), consistent with the majority of published work on this dataset.

CIC-IoT-2023 [17] captures IoT traffic from 105 heterogeneous devices with 33 attack types across 7 classes. With 46.7M records heavily skewed toward attacks (97.6%), we subsample to 1.3M records for a manageable benchmark, as a random split (no temporal ordering in the source data). Both the 1.3M subsample and full 46M dataset are published.

Dataset URLs anonymized for review; will be included in camera-ready.

3 Related Work

3.1 WNN-Based Intrusion Detection

The WiSARD [12] architecture pioneered RAM-based pattern recognition, and several works have applied WNNs to network security. Di Mauro et al. [7] experimentally compared twelve neural approaches for network intrusion management — including deep architectures, LVQ variants, and WiSARD — and found that WiSARD offered the best accuracy/time trade-off of all methods tested, identifying WNNs as strong candidates for near-real-time NIDS. Their evaluation, however, used vanilla WiSARD with fixed random connectivity on small subsampled data ($\sim 2\times 10^4$ flows per dataset), and reported neither false-positive rates nor temporal splits — precisely the gaps our work addresses.

The most directly related work is FWIW [21], which achieves 98.5% accuracy on UNSW-NB15 with a 192-byte model deployed on FPGA. FWIW uses Bloom filter-based neurons with H3 hashing and backpropagation training with random connectivity and one threshold mode. Our work differs in three ways: (1) we evaluate on both random and temporal splits, as random splits may allow some data leakage between train and test sets; (2) following Garcia and de Souto [10], who showed that global optimization of N-tuple connectivity outperforms random initialization, we apply a genetic algorithm to optimize neuron count; and (3) we evaluate seven threshold modes to identify which best predicts deployment performance.

BTHOWeN [23] introduced a *Gaussian* non-linear thermometer encoding for WNN classification — thresholds placed at equal-probability quantiles of a per-feature Gaussian fitted to the training data — achieving strong results on MNIST and eight UCI tabular datasets. We adopt thermometer encoding but replace the parametric Gaussian assumption with distributive (empirical-quantile) placement, which sets bin boundaries at equal-probability intervals of the actual training distribution. This avoids over-representing dense regions of noisy features (e.g., inter-arrival times with heavy tails) and ensures each thermometer bit carries roughly equal information content, improving robustness without requiring per-feature threshold optimization.

More recent WNN research has continued along this line: ULEEN [22] combines multi-pass training and submodel pruning for ultra-low-energy edge inference, and DWN [4] makes lookup tables differentiable for end-to-end gradient training. These works optimize encoding and training; none evolves the connectivity map itself, which is the focus of our approach.

3.2 Machine Learning for IDS

Traditional ML approaches (Random Forest, SVM, decision trees) achieve strong results on UNSW-NB15 [13] but require models of 10–50 MB, and the dataset's class overlap and imbalance make its standard partition intrinsically difficult [26]. Deep learning approaches (CNN, LSTM, GRU) push accuracy higher but typically require GPU hardware for inference [7], limiting deployment on power-constrained edge devices.

A critical methodological issue pervades the security ML literature: most published results use random train/test splits, reporting 97–99% accuracy that does not reflect deployment performance. Pendlebury et al. [16] demonstrated how such spatio-temporal experimental bias inflates malware-classification results; our own baseline measurements exhibit the same effect on UNSW-NB15, with Random Forest falling from 96.1% F1 on the random split (Section 6.1) to 85.2% on the temporal split (Section 6.1). For CICIDS2017, near-perfect random-split results dominate the literature, while temporal and concept-drift-aware evaluation remains rare [2,8]. Known labeling errors [9] further complicate comparisons.

3.3 FPGA-Based Network Security

FPGA acceleration for network security spans from packet-level inspection (Pigasus [25]) to ML inference (LogicNets [24], PolyLUT [3]). WNNs are particularly well-suited for FPGA deployment because RAM neurons map directly to FPGA lookup tables (LUTs) and block RAM (BRAM), requiring no multiply-accumulate units. FWIW [21] demonstrated this with a throughput of 740 million samples per second at 740 MHz on a Virtex UltraScale+. Our work extends this direction with evolved connectivity that can further reduce resource utilization by discovering sparser, more efficient neuron configurations.

4 Architecture

4.1 Single-Cluster Binary Discriminator

Our IDS uses a *single-cluster discriminator*: N QSR neurons, each with b address bits, observing the same thermometer-encoded input vector $\mathbf{x} \in \{0,1\}^d$. Each neuron i has a connectivity map $C_i = \{c_1^i, \ldots, c_b^i\}$ that selects b bits from $\mathbf{x}$ to form an address. The neuron outputs the value stored at that address in its 2^b-cell memory.

Training iterates over labeled examples: for each flow, each QSR neuron computes its address and writes the target label (0 for normal, 1 for attack) to the addressed cell using the QSR update rule (conflicting writes produce weak states, while confirming writes produce strong states).

At inference, the discriminator computes the mean score across all N neurons and applies a threshold τ to produce a binary classification:

$$\hat{y} = \begin{cases} 1 \text{ (attack)} & \text{if } \frac{1}{N}\sum_{i=1}^{N} \mathrm{mem}_i[\mathrm{addr}_i(\mathbf{x})] \geq \tau \\ 0 \text{ (normal)} & \text{otherwise} \end{cases} \tag{1}$$

This architecture is deliberately simple, a single cluster with one threshold, making the connectivity optimization problem tractable while still achieving competitive accuracy. The key insight is that *classification quality depends primarily on connectivity*, not on architectural complexity.

4.2 Phased Optimization Pipeline

We optimize the discriminator's network parameters (N, b, C) through a two-phase pipeline, where the output population of the first phase seeds the second.

Phase 1: Grid Search. Evaluation of (N, b) configurations (e.g., $N \in \{5, 10, \ldots, 500\}$, $b \in \{4, 8, \ldots, 34\}$) with random connectivity. Each configuration is trained on the full training set and evaluated using 5-fold cross-validation, then ranked by a fitness function. The top-K (N, b) configurations (default K=15) seed the initial population of P genomes (default P=50), distributed approximately evenly across the K architectures; for each selected (N, b) the grid-search genome is reused and the remaining quota is filled with fresh random-connectivity variants at the same architecture. This architectural diversity gives the subsequent GA multiple promising starting points rather than collapsing to a single configuration.

Phase 2: GA Neurons. A generational genetic algorithm refines model capacity starting from the grid-search population (Algorithm 1; hyperparameters in Table 1, identical across all datasets and runs).

Genome encoding. A genome is the triple $g = (\mathbf{N}, \mathbf{b}, \mathbf{C})$: per-cluster neuron counts $\mathbf{N}$ (a single cluster for the binary discriminator), per-*neuron* address widths $\mathbf{b} = (b_1, \ldots, b_{\sum N})$, and the connectivity map $\mathbf{C}$, stored as a flat integer list in which neuron j owns a contiguous slice of b_j input-bit indices. Genomes are variable-length: two genomes may differ in neuron count and per-neuron widths, and all operators handle this directly. Crucially, every operator moves *whole neurons* — a neuron's width and connection slice travel together, and surviving neurons keep their trained memory — so offspring remain in the phenotypic neighborhood of their parents; regenerating connections randomly would make "neighbors" behaviorally unrelated and destroy the search gradient. In this phase the widths of surviving neurons are inherited unchanged (widths are set by the grid phase); capacity and the connectivity of newly added neurons are what vary.

Parent selection. Size-3 tournament: three genomes are drawn uniformly at random and the one with the best fitness score (Equation 2) wins. The same score drives elitism and replacement, aligning all selection pressures.

Crossover (probability 0.7; otherwise the first parent is cloned). Uniform, position-wise, at neuron granularity: the child keeps its first parent's neuron count; for each neuron position present in both parents a fair coin decides which parent contributes that neuron (width and connections together); positions beyond the shorter parent inherit the longer parent's own neurons. Parents with different N are therefore always compatible.

Mutation (applied to every child; per-cluster probability 0.1). A mutated cluster's neuron count is perturbed by an integer drawn uniformly from $[-\delta_{\max}, \delta_{\max}]$, where $\delta_{\max} = \max(1, \mathrm{round}(0.1\,(N_{\min} + N_{\max})))$, clamped to $[N_{\min}, N_{\max}]$. Surviving neurons are copied verbatim; newly added neurons receive the cluster's modal width and fresh uniform-random connections.

Replacement (μ+λ with elitism). Each generation the top 20% of the population by fitness — deduplicated by architecture fingerprint, so identical genomes cannot monopolize elite slots — is carried over with cached evaluations. $\lceil P/0.75 \rceil$

Algorithm 1 GA-Neurons phase (one independent run)

$\mathcal{P} \leftarrow$ top-K grid architectures + random-connectivity variants ($|\mathcal{P}|=P$)
for $t = 1 \ldots G$ **do**
  $\mathcal{E} \leftarrow$ top 20% of $\mathcal{P}$ by Eq. 2, deduplicated ▷ elites, cached
  $\mathcal{O} \leftarrow \emptyset$
  while $|\mathcal{O}| < \lceil P/0.75 \rceil$ **do**
    $p_1, p_2 \leftarrow \text{tournament}_3(\mathcal{P}),\ \text{tournament}_3(\mathcal{P})$
    $c \leftarrow \text{crossover}(p_1, p_2)$ w.p. 0.7, else $\text{clone}(p_1)$
    $c \leftarrow \text{mutate}(c)$ ▷ perturb neuron counts
    train c; score by 5-fold CV ▷ batched, GPU
    if c viable **then** $\mathcal{O} \leftarrow \mathcal{O} \cup \{c\}$
    end if
  end while
  $\mathcal{O} \leftarrow$ top P of $\mathcal{O}$ by Eq. 2
  $\mathcal{P} \leftarrow$ top P of $\mathcal{E} \cup \mathcal{O}$ ▷ $\mu+\lambda$ truncation
  every 10 gens: stop after 5 checks without $\geq$0.05% gain
end for

offspring are bred and evaluated, ranked, and the top P retained (the 0.75 selectivity percentile); offspring must also exceed a progressive minimum-accuracy viability threshold, filtering degenerate architectures before they consume evaluation budget. Elites and offspring are then pooled and the best P by fitness form the next generation.

Evaluation. Every new genome is trained from scratch and scored by 5-fold cross-validation *within the 80% training partition* (train on four folds, score the held-out fifth, rotate, average); the 20% held-out test set is never touched during search (Section 5). All candidate evaluations in a generation are dispatched as one batch to the Rust/Metal accelerator.

Termination. At most G=250 generations, with early stopping: every 10 generations, if the best fitness has not improved by at least 0.05% since the previous check, a patience counter increments, and the search stops after 5 consecutive unimproved checks (runs typically terminate well before the cap). All stochastic choices derive from a single per-run seed, making each run reproducible.

This phase discovers the optimal model capacity per dataset— pruning neurons on CICIDS2017 and UNSW-NB15, expanding neurons and bits on the higher-volume CIC-IoT-2023—while consistently improving accuracy over the grid-search baseline (Section 6.2).

Fitness function. IDS metrics (F1, FPR, cross-entropy, accuracy) differ in scale, making weighted sums sensitive to normalization. Both phases rank genomes per metric within the population, then combine ranks via a weighted harmonic mean:

$$\text{fitness}(g) = \frac{w_{CE} + w_{F1} + w_{FPR} + w_{Acc}}{\frac{w_{CE}}{r_{CE}(g)} + \frac{w_{F1}}{r_{F1}(g)} + \frac{w_{FPR}}{r_{FPR}(g)} + \frac{w_{Acc}}{r_{Acc}(g)}} \tag{2}$$

where $r_m(g)$ is genome g's rank for metric m (lower rank = better). The CICIDS2017 runs use w_{CE}=0.35, w_{Acc}=0.30, w_{F1}=0.30, w_{FPR}=0.05. For UNSW-NB15 temporal and CIC-IoT-2023, we adopt w_{CE}=0.1, w_{F1}=0.35, w_{FPR}=0.35,

Table 1. GA hyperparameters (identical across all datasets and runs).

Population size P	50
Max generations G	250
Tournament size	3
Crossover probability	0.7
Mutation probability (per cluster)	0.1
Elitism fraction	20%
Offspring selectivity percentile	0.75
Replacement	$\mu+\lambda$ truncation
Early-stop check interval	10 generations
Early-stop patience	5 checks
Min. improvement per check	0.05%
Fitness evaluation	5-fold CV, 80% train only

w_{Acc}=0.2, balancing F1 and FPR equally (Section 6.1). The refinement was driven by empirical observation that the original FPR-heavy weights caused the GA to over-optimize for low FPR at the expense of F1; equal F1/FPR weighting produces genomes with a better balanced operating point across threshold modes.

4.3 GPU-Accelerated Optimization

Population-based optimization requires evaluating hundreds of genome configurations per generation across hundreds of generations. We implement a Rust accelerator with GPU support that parallelizes both training and evaluation. A complete two-phase run (grid search + GA neurons) completes in ∼5 minutes per seed on UNSW-NB15, enabling 100+ statistically independent runs per dataset.

5 Evaluation Methodology

5.1 Temporal and Random Evaluation Protocols

Evaluation protocol is a critical factor in IDS benchmarking. Random train/test splits allow training and test sets to share temporal characteristics (and, in datasets with duplicate records, near-identical flows), which can lead to higher reported accuracy. Temporal splits, where training data precedes test data chronologically, better simulate deployment conditions where the model must generalize to future traffic patterns.

We evaluate under the temporal protocol for UNSW-NB15 and random splits for UNSW-NB15, CICIDS2017, and CIC-IoT-2023. All splits are published on HuggingFace for reproducibility:

– **Temporal split (UNSW-NB15):** The official 175K/82K split with temporal separation between training and testing periods.

- **Random splits (all datasets):** Stratified 80/20 splits on UNSW-NB15 (1.27M/317K), CICIDS2017 (2.3M/566K), and CIC-IoT-2023 (1.07M/268K; 1.3M subsample).
- **Full-scale CIC-IoT-2023:** The full 46.7M-record dataset with random 80/20 split (37.4M/9.3M), used for direct comparison with prior published baselines (Section 6.1).

5.2 Seven-Mode Threshold Calibration

Binary IDS classifiers produce a continuous score that must be thresholded to yield a classification. The choice of threshold dramatically affects F1 and FPR, yet most published work reports results under a single (a fixed 0.5, or sometimes unspecified) threshold mode. We evaluate every genome under seven calibration strategies to expose this sensitivity:

1. **Train-calibrated (train_cal):** Threshold swept on training-set scores to maximize the same fitness function used by the GA.
2. **Fixed 0.5:** No calibration; the midpoint threshold (distribution-agnostic baseline).
3. **Platt scaling** [18]: Logistic regression on held-out scores, $P(\text{attack}) = \sigma(as + b)$.
4. **Beta calibration** [11]: Three-parameter logit model, more flexible than Platt for asymmetric distributions.
5. **Empirical:** Histogram-based threshold selected by sweeping over a discrete grid of held-out scores (simple non-parametric baseline).
6. **Empirical cumulative:** Threshold chosen at the score percentile of the cumulative distribution that matches the training positive rate.
7. **Validation (val_cal):** Threshold swept on the held-out validation set (not used during training or GA selection). This represents an upper bound on achievable performance, not available in deployment, but quantifies the calibration gap.

The gap between validation and train-calibrated performance quantifies how much accuracy is "left on the table" by imperfect threshold selection, a finding we report for each dataset and configuration.

Fitness-aligned threshold optimization. The train_cal mode sweeps the threshold to maximize the *same* weighted-fitness function (Equation 2) that the GA uses for genome selection, rather than maximizing F1 alone. This alignment is methodologically critical: when the threshold sweep optimizes F1 but the GA selects on a fitness combining CE, F1, FPR, and accuracy, the GA receives misleading FPR signals and discards genomes that would have been strong under the actual fitness criterion. The co-optimization ensures that architecture search and threshold selection work together rather than at cross purposes.

5.3 Statistical Protocol

All cohorts target n=100 independent flows per (dataset, configuration) tuple, providing uniform statistical treatment across the four datasets evaluated. We report cohort means with standard deviations alongside the best individual genome on each Pareto dimension; per-dataset cross-seed standard deviations appear in the corresponding result tables. With n=100 and a conservative typical $\sigma \approx 2\%$, the 95% confidence interval for the cohort mean is ±0.39%, sufficient to distinguish meaningful differences between configurations; the per-dataset ± values reported in result tables are substantially tighter than this bound on most configurations.

Tree-based baselines (Random Forest, XGBoost) are reported as single values with fixed hyperparameters and default library seeds. These models are near-deterministic under fixed configuration, so cross-seed variance is much smaller than the evolutionary variance of our population-based search, and the performance gaps reported in Tables 2–5 are robust to baseline seed choice.

K-fold cross-validation (k=5) is applied *every generation* during evolution within both phases: the training set is partitioned into 5 folds, and each genome is trained and evaluated on all 5 train/validation splits every generation. The averaged fitness across all 5 folds determines selection, preventing the GA from overfitting to a particular data partition and providing more robust genome ranking. Early stopping terminates evolution when the best K-fold fitness does not improve by at least 0.05% for 5 consecutive check intervals (every 10 generations), avoiding unnecessary computation and further guarding against overfitting.

All reported results use the *best-fitness genome from the final optimization phase*, evaluated by the full-dataset evaluator (175K train → 82K test for UNSW-NB15) with all seven threshold modes. Each flow contributes one data point to the aggregate statistics, ensuring that reported means reflect the full distribution of outcomes.

6 Experimental Results

6.1 Dataset Results

UNSW-NB15 Temporal Results. Table 2 reports results from 30 independent seeds on the UNSW-NB15 temporal split (16-bit thermometer, top-20 RF features, population size 50) using balanced fitness weights (w_{F1}=0.35, w_{FPR}=0.35, w_{CE}=0.1, w_{Acc}=0.2) and fitness-aligned threshold optimization (Section 5). Random Forest and XGBoost baselines were measured on the identical 80/20 temporal partition, on both raw top-20 numeric features and the matched 16-bit thermometer encoding (Table 2).

The validation-calibrated threshold achieves 88.87% ± 0.23% F1 at 8.77% FPR, outperforming Random Forest (86.05% F1, 25.41% FPR, at its strongest 16-bit thermometer encoding) by 2.82 pp F1 at roughly one-third the FPR, and outperforming XGBoost (84.89% F1, 28.57% FPR) by 3.98 pp F1 at less than one-third the FPR; Table 2 reports both raw and thermometer baselines. The

0.23 pp standard deviation across 30 independent seeds indicates that the GA's discovery of competitive genomes is highly reproducible. The best single genome reaches 89.34% F1 at 9.76% FPR; within the same evolutionary population we identify a 5-neuron Pareto operating point at 87.88% F1 and 6.94% FPR with approximately 2 KB of model memory (single-genome rows in Table 2). This 5-neuron variant beats Random Forest by 1.83 pp F1 and 18.47 pp FPR at roughly 138,000× smaller model size.

The train-calibrated threshold reaches 83.97% ± 0.33% F1 at 28.70% FPR. The roughly 5 pp F1 gap and 20 pp FPR gap between train_cal and val_cal modes on this split is substantially larger than on either the random-split partition or the other datasets reported in this paper. It reflects the temporal distribution shift between training traffic (earlier flows) and test traffic (later flows), which moves the optimal decision threshold by an amount that the training-set calibration cannot anticipate. Validation calibration samples the held-out distribution at deployment time to recover the target operating point. We report both modes together to expose this gap rather than silently use the more favorable calibration. Two intermediate modes refine the picture. The empirical-cumulative threshold—which matches the training positive rate against the test-time score distribution and therefore needs no held-out labels—recovers most of the gap *deployably*, reaching 88.45% F1 at 12.40% FPR (less than half the train-calibrated FPR) with tight variance (±0.34% F1). The histogram-based empirical threshold attains the lowest cohort FPR (5.57% at 87.18% F1), but with high variance (±4.51% FPR) that reflects its sensitivity to score-distribution noise and makes it a less reliable operating point.

The cohort row reports the best-CE-selected GA Neurons genome rather than the best-fitness selection. In this cohort, the best-fitness and best-F1 selections collapse to the same evolutionary endpoint (identical genome hashes per seed), while the best-CE selection identifies a distinct, larger architecture (309 ± 107 vs. 131 ± 158 neurons) that dominates the best-fitness selection on *both* F1 and FPR at val_cal (88.86 vs. 85.88 F1, 8.78 vs. 17.05 FPR). The full breakdown across all five genome selections is reported in Appendix A; we elected the best-CE row for the body table because it is the dominant Pareto point for deployment under the corrected pipeline.

UNSW-NB15 Random Results. Table 3 reports results from 30 independent seeds on the UNSW-NB15 random 80/20 split (64-bit thermometer, top-20 RF features, population size 50) using the same balanced fitness weights as the temporal split (w_{F1}=0.35, w_{FPR}=0.35, w_{CE}=0.1, w_{Acc}=0.2) and fitness-aligned threshold optimization (Section 5), evaluated under the order-independent training pipeline. Random Forest and XGBoost baselines were measured on the identical 80/20 random partition, on both raw numeric features and the 64-bit thermometer encoding (Table 3); here raw is the trees' stronger configuration.

The validation-calibrated threshold achieves **93.56**% ± 0.12% F1 at 1.07% FPR. On this comparatively easy random split the tree ensembles are very strong (Random Forest 96.06% F1 on raw features, 95.46% on thermometer bits), so

Table 2. UNSW-NB15 temporal split (30 independent seeds, 16-bit thermometer, balanced fitness weights w_{F1}=0.35, w_{FPR}=0.35, w_{CE}=0.10, w_{Acc}=0.20). Cohort rows report the best-CE-selected GA Neurons genome at each threshold mode (mean ± sample std, n=30); see Section 6.1 for the genome-type rationale. Cohort genome size for the best-CE selection: 309 ± 107 neurons, 33 ± 1 bit-address. Single-genome rows report the best discovered configuration for each Pareto operating point. Full seven-mode × five genome-type breakdown in Appendix A.

	F1 (%)	FPR (%)	Acc (%)	Size	Latency	Power
Cohort distribution (n=30 seeds)						
Train-cal	83.97±0.33	28.70±1.00	84.65±0.27	~1.3 MB^{M}	—	—
Fixed 0.5	84.40±0.49	27.35±1.47	85.00±0.42	~1.3 MB^{M}	—	—
Empirical	87.18±1.14	5.57±4.51	87.22±1.11	~1.3 MB^{M}	—	—
Empirical cumul.	88.45±0.34	12.40±1.04	88.56±0.33	~1.3 MB^{M}	—	—
Val_cal	**88.87**±0.23	8.77±1.14	88.92±0.23	~1.3 MB^{M}	—	—
Single-genome Pareto operating points (val_cal)						
Best F1‡	**89.34**	9.76	89.42	~1.3 MB	—	—
Edge / FPGA (5 neurons)†	87.88	6.94	—	~2 KB	—	—
Low-FPR (16 neurons)†	87.70	**4.63**	—	~5 KB	—	—
Measured baselines (identical partition; raw + 16-bit thermometer)^{B}						
Random Forest (raw)	85.18	27.73	85.82	138 MB	14.0 μs^{C}	—
Random Forest (16b therm)	86.05	25.41	86.39	138 MB	—	—
XGBoost (raw)	84.93	28.68	85.63	0.27 MB	0.4 μs^{C}	—
XGBoost (16b therm)	84.89	28.57	85.34	0.27 MB	—	—

‡Best single-genome F1 across the n=30 cohort. †Pareto operating points from the same evolutionary populations: the 5-neuron variant comes from the same flow that produced the F1-best genome and fits the Zynq Z-7020 internal BRAM with 99% headroom; the 16-neuron variant minimizes FPR within a single-LUT footprint envelope. ^{M}Cohort genome memory size estimate from per-flow neuron count and sparse address coverage. Vivado synthesis (LUTs, latency, power) for the corrected cohort is in progress; resource utilization for the 5-neuron edge variant is projected at sub-1,000 LUTs based on architecture-equivalent prior synthesis. ^{C}CPU (Apple M4 Max) inference, *batch-amortized*: total `predict()` time over the held-out set divided by the sample count — a best-case lower bound that *favors* the CPU baselines (true single-sample latency is higher). The WNN *Latency* and *Power* columns await Vivado synthesis of the corrected n=30 cohort (in progress; see ^{M}). ^{B}Baselines are reported on *both* raw numeric top-20 features (the trees' natural format) and the matched 16-bit thermometer encoding (the WNN's input representation); thermometer bits tend to help the trees by axis-aligning the decision boundaries, so we report the WNN's margin against the stronger (thermometer) configuration. Random Forest: scikit-learn 100 trees, max_depth=None; XGBoost: 100 trees, max_depth=6, lr=0.1.

the WNN trails Random Forest by ~2.5 pp F1; its advantage here is efficiency—ensemble-class accuracy within a few points at 10^4–$10^6\times$ smaller model size. The 0.12 pp standard deviation across 30 independent seeds confirms the reproducibility observed on the temporal split. The best single genome reaches **93.92**% F1 at 0.68% FPR, and the same evolutionary populations expose a minimum-FPR operating point at 0.07% FPR for deployment-oriented operation.

In contrast to the temporal split, the train_cal and val_cal thresholds nearly coincide on the random partition: with matched train/test distributions there is little calibration headroom, so both modes report essentially the same operating point. The ∼5 pp F1 / ∼20 pp FPR train_cal–val_cal gap on the temporal split (Section 6.1) is specific to its temporal distribution shift and does not arise here.

Table 3. UNSW-NB15 random split (30 independent seeds, 64-bit thermometer, balanced fitness weights w_{F1}=0.35, w_{FPR}=0.35, w_{CE}=0.10, w_{Acc}=0.20). Cohort rows report the [genome-type TBD]-selected GA Neurons genome at each threshold mode (mean ± sample std, n=30); single-genome rows report the best discovered configuration for each Pareto operating point. Full seven-mode × five genome-type breakdown in Appendix A.

	F1 (%)	FPR (%)	Acc (%)	Size	Latency	Power
Cohort distribution (n=30 seeds, best-fitness GA genome)						
Train-cal	93.56±0.12	1.07±0.11	98.94±0.04	∼1 MB^M	—	—
Fixed 0.5	92.13±1.44	1.41±0.32	98.64±0.30	∼1 MB^M	—	—
Val_cal	**93.56**±0.12	1.07±0.11	98.94±0.04	∼1 MB^M	—	—
Single-genome Pareto operating points						
Best F1‡	**93.92**	0.68	99.07	∼1 MB	—	—
Best FPR†	83.79	**0.07**	98.14	—	—	—
Measured baselines (identical partition; raw + 64-bit thermometer)^B						
Random Forest (raw)	**96.06**	0.30	99.42	138 MB	14.0 μs^C	—
Random Forest (64b therm)	95.46	0.34	99.34	138 MB	—	—
XGBoost (raw)	95.54	0.34	99.35	0.27 MB	0.4 μs^C	—
XGBoost (64b therm)	95.35	0.29	99.33	0.27 MB	—	—

‡Best single-genome F1 across the n=30 cohort. †Low-FPR Pareto operating point from the same evolutionary populations. ^{M}Cohort genome memory size estimate from per-flow neuron count and sparse address coverage; Vivado synthesis (LUTs, latency, power) for the corrected 64-bit cohort is in progress (the prior 16-bit dense 47-neuron synthesis and the FWIW comparison are retained in the source for re-integration). ^{C}CPU (Apple M4 Max) inference, *batch-amortized*: total `predict()` time over the held-out set divided by the sample count — a best-case lower bound that *favors* the CPU baselines (true single-sample latency is higher). The RF/XGBoost latencies are the measured top-20-feature, 100-tree model class (identical hyperparameters and feature set to the temporal split, so the partition-independent per-sample cost is the same); the WNN *Latency* and *Power* columns await Vivado synthesis of the corrected 64-bit cohort (in progress; see ^{M}). ^{B}RF/XGBoost reported on both raw numeric top-20 features and the matched 64-bit thermometer encoding (the WNN's input); on this split raw is the trees' stronger configuration (RF 96.06 vs 95.46), so we compare against raw. Random Forest: scikit-learn 100 trees, max_depth=None; XGBoost: 100 trees, max_depth=6, lr=0.1.

CICIDS2017 Random Results. Table 4 reports results from 30 independent seeds on CICIDS2017 (96-bit thermometer, top-20 RF features, random

80/20 split, population size 50) using the CE-leading fitness weights (w_{CE}=0.35, w_{Acc}=0.30, w_{F1}=0.30, w_{FPR}=0.05) and fitness-aligned threshold optimization (Section 5), evaluated under the order-independent training pipeline. Random Forest and XGBoost baselines were measured on the identical random partition, on both raw numeric features and the 96-bit thermometer encoding (Table 4); here the thermometer encoding is the trees' stronger configuration.

Performance on this near-separable random split is very high—similar to CIC-IoT-2023 and well above the UNSW-NB15 temporal split, as random splits let similar network sessions appear in both train and test sets and the identical class distributions eliminate temporal distribution shift. The validation-calibrated threshold achieves **99.55**% ± 0.05% F1 at 0.12% FPR, with the train-calibrated threshold coinciding (with matched train/test distributions there is little calibration headroom, as on the UNSW random split). Among individual genomes, the best F1 reaches **99.64**% F1 / 0.08% FPR / 99.77% Acc—essentially matching XGBoost on raw features (99.65% F1, 99.78% Acc) while *halving* its false-positive rate (0.08 vs 0.12%) and matching Random Forest's raw-feature FPR (0.08%)—and the same evolutionary populations expose a minimum-FPR operating point at **99.37**% F1 / **0.053**% FPR / 99.61% Acc for deployment-oriented operation, at 10^4–$10^6\times$ smaller model size than the tree ensembles.

CIC-IoT-2023 Results. The CIC-IoT-2023 results reported in this section use three methodology refinements adopted post-submission: a 96-bit thermometer encoding per feature (motivated by the encoding-width sweep in Section 7), an enlarged architecture search ceiling ($N_{\max}$=250 neurons, $b_{\max}$=100 bits per neuron), and an order-independent vote-tally implementation of the QSR update rule (Section 2). UNSW-NB15 and CICIDS2017 results above retain the submitted-paper methodology.

We evaluate on CIC-IoT-2023 using two complementary protocols: a 30-run statistical batch on a 1.14M-train random subsample (96-bit thermometer encoding per feature, top-20 RF features, population size 50) using CE-heavy fitness weights (w_{CE}=0.70, w_{Acc}=0.10, w_{F1}=0.15, w_{FPR}=0.05) and fitness-aligned threshold optimization (Section 5), and a single-genome evaluation on the full 46.7M-record dataset for direct comparison with prior published baselines. This cohort supersedes an earlier 100-run batch whose runs predated fixes to the order-independent trainer and the QUAD cell semantics; the refreshed numbers are within noise of the earlier ones at unconstrained operating points and strictly better in the low-FPR band.

The CE-heavy weights de-emphasize the FPR component of the fitness function. We found empirically that high w_{FPR} at low bits-per-neuron drives the GA toward overfit-to-training genomes with degraded test FPR (*best-FPR paradox*, Section 7); demoting w_{FPR} to 0.05 rehabilitates this failure mode. With max neuron count 250 and max bits per neuron 100, the GA converges on b=64±2 bits per neuron (most seeds select 64 bits, with rare excursions to neighboring widths) at $\overline{n}$=228±18 neurons across seeds. The bit-width and neuron count are reproducible to within a few percent across independent runs, and cross-seed

Table 4. CICIDS2017 random split (30 independent seeds, 96-bit thermometer, CE-leading fitness weights w_{CE}=0.35, w_{Acc}=0.30, w_{F1}=0.30, w_{FPR}=0.05). Cohort rows report the best-fitness GA Neurons genome at each threshold mode (mean ± sample std, n=30); single-genome rows report the best discovered configuration for each Pareto operating point. Full seven-mode × five genome-type breakdown in Appendix A.

	F1 (%)	FPR (%)	Acc (%)	Size	Latency	Power
Cohort distribution (n=30 seeds, best-fitness GA genome)						
Train-cal	**99.55**±0.05	0.12±0.05	**99.72**±0.03	∼1 MB^M	—	—
Fixed 0.5	99.17±0.12	0.50±0.10	99.47±0.08	∼1 MB^M	—	—
Val_cal	**99.55**±0.05	0.12±0.05	**99.72**±0.03	∼1 MB^M	—	—
Single-genome Pareto operating points						
Best F1‡	**99.64**	0.08	**99.77**	∼1 MB	—	—
Best FPR†	99.37	**0.053**	99.61	—	—	—
Measured baselines (identical partition; raw + 96-bit thermometer)^B						
Random Forest (raw)	99.74	0.08	99.84	6.3 MB	13.4 µs^C	—
Random Forest (96b therm)	**99.78**	**0.07**	**99.86**	6.3 MB	—	—
XGBoost (raw)	99.65	0.12	99.78	0.3 MB	0.4 µs^C	—
XGBoost (96b therm)	99.68	0.09	99.80	0.3 MB	—	—

‡Best single-genome F1 across the n=30 cohort (genome e9641). †Low-FPR Pareto operating point (genome e9639) from the same evolutionary populations. ^{M}Cohort genome memory estimate from per-flow neuron count and sparse address coverage; 96-bit Vivado synthesis (LUTs, latency, power) for the FPGA rows is in progress. ^{C}CPU (Apple M4 Max) inference, *batch-amortized*: total `predict()` time over the held-out set divided by the sample count — a best-case lower bound that *favors* the CPU baselines (true single-sample latency is higher). The WNN *Latency* and *Power* columns await Vivado synthesis of the corrected n=30 cohort (in progress; see ^{M}). ^{B}RF/XGBoost reported on both raw numeric top-20 features and the matched 96-bit thermometer encoding (the WNN's input); thermometer is marginally stronger here (RF 99.78 vs 99.74). Random Forest: scikit-learn 100 trees, max_depth=None; XGBoost: 100 trees, max_depth=6, lr=0.1.

F1 standard deviation on calibrated thresholds is 0.15–0.17 pp—tight enough to make the cohort mean a stable estimator of the architecture's true performance.

Comparison with prior baselines. The GA Neurons mean at train_cal threshold (F1= 92.81%, Acc= 96.42%) exceeds AdaBoost (top-20): F1= 91.61%, Acc= 95.86%, at lower FPR (8.57% vs. 11.77%) and matches XGBoost (93.36% F1, 96.81% Acc) on F1 within 0.55 pp at FPR lower by 3.08 pp (8.57% vs. 11.65%). The GA Neurons fixed_05 threshold then drives FPR below 1% (0.96±0.11%) at F1= 86.43±0.88% — a sub-1% mean-FPR operating point unreachable by tree ensembles at any threshold (Random Forest's lowest FPR with F1>80% is 9.77%, XGBoost's is 11.65%). Per-genome Pareto extremes span the calibration spectrum: at the fixed_05 threshold a best-FPR seed reaches F1=88.48% at FPR=1.06% (ultra-low-FPR cascade-front-stage deployment; the cohort's lowest FPR point is F1=88.27% at FPR=0.72%), and at the empirical_cumulative threshold a best-FPR seed lands F1=92.73% at FPR=4.86% —

Table 5. CIC-IoT-2023 1.14M random subsample (96-bit thermometer encoding per feature, CE-heavy fitness weights), canonical TOP20 features re-derived on neto-full. GA converges on b=64±2 bits and $\overline{n}$=228±18 neurons across seeds (most seeds select 64 bits per neuron, with rare excursions to neighboring widths). Statistics from 30 of 30 runs of the cohort (complete), as of 03/07/2026. Per-genome Pareto points reported on the same cohort; hardware sizing (Size/Latency/Power) was Vivado-measured on the superseded cohort's identically shaped GA-converged 64-bit genomes (footnote E); re-synthesis of the refreshed winners is pending.

	F1 (%)	FPR (%)	Acc (%)	Size	Latency	Power
Best F1‡	93.17	8.07	96.60	—	—	—
Best F1 (FPR<10%)‡	93.17	8.07	96.60	—	—	—
Best F1 (FPR<6%)‡	93.01	5.89	96.45	—	—	—
Best F1 (FPR<5%)‡	92.73	4.86	96.26	—	—	—
Best F1 (FPR<4%)‡	88.48	1.06	93.46	—	—	—
Best Acc‡	93.17	8.37	96.61	—	—	—
Grid Search (train_cal)	90.16±0.16	15.29±0.67	95.19±0.11	—	—	—
GA Neurons (train_cal)	**92.81**±0.17	8.57±0.67	**96.42**±0.08	—	—	—
GA Neurons (fixed_05)	86.43±0.88	**0.96**±0.11	92.04±0.62	—	—	—
Random Forest (96b therm)	94.43	9.77	97.33	~1 GB	1.4 μs^C	—
Random Forest (raw)	85.53	25.18	92.71	~0.9 GB	—	—
XGBoost (96b therm)	93.36	11.65	96.81	0.3 MB	0.1 μs^C	—
XGBoost (raw)	84.13	28.34	92.07	0.3 MB	—	—
AdaBoost (top-20)	91.61	11.77	95.86	30 KB	0.8 μs^C	—
AdaBoost (all 46 feat)	90.86	9.36	95.33	30 KB	0.9 μs^C	—
Perceptron	66.58	60.09	84.60	<10 KB	6 ns^C	—

‡Best single genome. ^{C}CPU (Apple M4 Max), *batch-amortized*: per-sample latency is the total `predict()` time over the held-out set divided by the sample count. True single-sample (streaming) latency is higher — batching amortizes SIMD, cache-residency, and per-call overhead — so this is a best-case lower bound that *favors* the CPU baselines, whereas the WNN FPGA figures are true single-sample hardware latency. ^{E}*External-DRAM-backed.* Hardware figures were Vivado-measured on the superseded cohort's identically shaped winners (211–247 neurons × 64–67 bit, 12.7–13.3M sorted entries: ~13.6k RAMB18, 108.7–114.6 MB, ~72–84 μs, 165–303 mW under the model below); the refreshed cohort converges to the same family (195–250 neurons × 59–64 bit) and re-synthesis of its winners is pending. Such sorted tables far exceed any FPGA's on-chip BRAM, so they live in external DDR3. *Size* = model storage. *Latency* = single-DDR3-port contention (cycle-accurate arbiter model: 17 dependent binary-search probes/neuron, SERVICE = 4 / read-latency = 30 cycles @ 200 MHz). *Power* = on-chip search logic only (N binary-search FSMs, $F_{max} \approx$ 74–77 MHz, zero BRAM); DDR3 PHY+DRAM power excluded. RF/XGBoost: CPU. Baseline sizes via `joblib.dump`. RF/XGBoost are reported on both the matched 96-bit per-feature thermometer encoding (the WNN's input) and raw numeric features; thermometer bits substantially help the trees on this dataset (RF +8.9 pp F1), so we compare against that stronger configuration.

matching XGBoost on F1 within 0.63 pp at FPR lower by 6.79 pp. The cohort's overall best individual genome (val_cal threshold) reaches F1=93.17%, FPR=8.07%, Acc=96.60% — matching XGBoost on F1 within 0.19 pp at FPR lower by 3.58 pp and Acc within 0.21 pp. This is done with a ∼228×64 WNN that is, at FPGA deployment, orders of magnitude smaller and faster than the XGBoost CPU baseline.

Full 46M dataset. To compare with prior work that reports on the full 46.7M-record CIC-IoT-2023, we evaluate small-genome architectures on the complete canonical Neto distribution (`lacg030175/CIC-IoT-2023-neto-full`, sourced from `akashdogra/ciciot23csv`; 37.3M train / 9.3M test test+val merged, random 80/20). Architectures were selected from the 1.3M-subsample search and transferred to 46M without further GA refinement, isolating the effect of training-data volume from architecture-search effects.

Feature ranking for thermometer-encoded WNNs. For the Micro and Small WNN rows in Table 6, we compare two feature-selection criteria: the standard Random Forest importance ranking (TOP20-RF), used by all tree-ensemble baselines, and a quantization-aware ranking we propose (TOP20-MI8b), computed as mutual information between each feature's 8-bit thermometer-quantized representation and the binary attack label. The two rankings overlap on 18 of 20 features. However, the substitutions matter architecture-by-architecture: TOP20-RF yields the smallest deployable detector at sub-5% FPR (300n×4b, 1.2 KB, 86.81% F1, 4.38% FPR), while TOP20-MI8b yields the highest-F1 sub-10% FPR detector (300n×8b, 18.8 KB, 88.35% F1, 6.60% FPR) and improves three additional Pareto points (100n×4b, 200n×4b, 400n×8b). Each table row reports the better-performing ranking, marked ^{R} or ^{M}. This is consistent with the model-class divergence observed in baseline reruns: when both rankings are evaluated, RF and XGBoost see comparable gains from the canonical TOP20 (because tree splits handle continuous features natively), while the WNN's optimal feature subset depends on the per-neuron address width. Table 6 groups the transferred results into two deployment tiers — micro (400 B to 1.2 KB) and small (18–25 KB) — and compares against the direct 46M search (Search(46M) rows) and the baselines reported by Neto et al. [17]. At matched FPR (∼1.6%), the direct search improves F1 by 4.75 pp over the best transferred Peak genome (89.58% vs. 84.83%) — an order-of-magnitude larger gain than the 0.47 pp seen in pre-refinement runs, attributable to the order-independent vote-tally training rule adopted post-submission (Section 2) which removes the example-order noise that bottlenecked prior search. At best-F1 (92.18%), the direct search closes ∼5 pp of the gap to XGBoost (96.03%) while remaining 1.34× deeper on FPR (6.73% vs. 5.01%); at best-FPR (0.71%), the WNN trades 9.3 pp F1 for a ∼7× FPR reduction below XGBoost — a meaningful operating point for high-cost-of-false-positive SOC deployments where alert volume drives analyst workload. The sub-1% FPR figure also beats RF's 3.46% by ∼5× at substantially smaller footprint.

6.2 Phase Progression Analysis

Our two-phase pipeline first performs a grid search over neuron counts and address widths, then refines the best genome with a genetic algorithm that mutates

Table 6. Full 46M CIC-IoT-2023 (random 80/20) on the canonical Neto distribution (`lacg030175/CIC-IoT-2023-neto-full`). Micro/Small: per-architecture best between TOP20-RF (Random Forest importance) and TOP20-MI8b (quantization-aware mutual information) feature rankings. Peak rows reflect prior dataset mirrors; Search(46M) reports the post-submission 250n×(60–64)b architectures from flows 2747+2748 (n=2 seeds, completed 27/05/2026); FPGA columns marked TBD^V are pending Vivado synth from the 250n×mix-bits exports.

Tier	Method	F1 (%)	FPR (%)	Acc (%)	Size	LUTs	Latency	Power
Micro	Ours (100n × 4b)^M	85.27	7.95	98.27	400 B	81 (0.15%)	11 ns^D	103 mW
	Ours (200n × 4b)^M	86.13	12.96	98.49	800 B	154 (0.3%)	13 ns^D	103 mW
	Ours (300n × 4b)^R	**86.81**	**4.38**	**98.45**	**1.2 KB**	**229 (0.4%)**	**13 ns**^{D}	**103 mW**
Small	**Ours (300n × 8b)**^{M}	**88.35**	**6.60**	**98.71**	**18.8 KB**	**229 (0.4%)**	**13 ns**^{D}	**103 mW**
	Ours (400n × 8b)^M	87.37	8.76	98.60	25 KB	304 (0.6%)	15 ns^D	103 mW
Peak	Ours (96n × 32b)	84.36	1.61	97.59	3.4 MB	9,492 (17.8%)	603 ns^S	312 mW
	Ours (198n × 32b)	84.68	2.00	97.67	6.3 MB	19,468 (36.6%)	680 ns^S	521 mW
	Ours (245n × 32b)	84.79	1.59	97.68	7.8 MB	23,877 (44.9%)	761 ns^S	620 mW
	Ours (500n × 34b)	84.73	1.91	97.68	18.0 MB	50,527 (95.0%)	845 ns^S	1.16 W
Search (46M)	Best F1‡ (250n × 60–64b)	**92.18**	6.73	**99.21**	TBD^V	TBD^V	TBD^V	TBD^V
	Best FPR‡ (250n × 60–64b)	88.34	**0.71**	98.64	TBD^V	TBD^V	TBD^V	TBD^V
	Matched FPR$^\beta$ (250n × 60–64b)	89.58	1.65	98.83	TBD^V	TBD^V	TBD^V	TBD^V
Our Baseline (46M)	Random Forest	97.61	3.46	99.78	1.0 GB	—	159 ns^C	—
	XGBoost	96.03	5.01	99.63	400 KB	—	58 ns^C	—
	AdaBoost (top-20)	93.13	11.88	99.36	33 KB	—	544 ns^C	—
	AdaBoost (all 46 feat)	93.12	11.88	99.36	33 KB	—	696 ns^C	—
	Perceptron	82.06	46.25	98.63	2.4 KB	—	59 ns^C	—
Neto et al.† (46M)	DNN	94.03†	—	99.44	∼5 MB	—	—	—
	AdaBoost	95.63†	—	99.59	∼10 MB	—	—	—
	Random Forest	96.53†	—	99.68	∼50 MB	—	—	—

Micro/Small: train_cal threshold, best-fitness genome (single seed). ^{R}TOP20-RF: top-20 by Random Forest importance (neto-full). ^{M}TOP20-MI8b: top-20 by mutual information of 8-bit thermometer-quantized values with the label (see "Feature ranking for thermometer-encoded WNNs"). ‡Best genome across threshold modes. $^\beta$Beta-calibrated to the prior Peak-row FPR target (∼1.6%). ^{V}Pending Vivado synth from the 250n×mix-bits exports (in progress). ^{D}Dense $O(1)$ lookup, Z-7020. ^{S}Sparse $O(\log n)$: 3 FSM cycles/probe, 49–58 cycles (Vivado xsim), latency = cycles/F_{max}; these exceed the Z-7020's 0.63 MB BRAM so the on-chip figure presumes a larger device, DRAM-backed single-port contention gives ∼20–90 μs. ^{C}CPU (M4 Max), *batch-amortized* (`predict()` time / samples): a best-case lower bound *favoring* the CPU baselines; WNN FPGA figures are true single-sample. †Neto et al.: F1/Acc as reported in [17] (averaging unspecified there), StandardScaler + different split.

neuron count while preserving evolved connections. Table 7 compares the two phases across all three datasets (val_cal threshold, best-fitness genome).

The GA phase consistently improves F1 over the grid-search baseline; the neuron-count direction is dataset-dependent. On CICIDS2017, the GA reduces neurons by 42% (343→198) while improving F1 by 0.11 pp and halving variance (±0.76% → ±0.48%), indicating that neuron-count optimization acts as a regularizer. On the harder UNSW-NB15 temporal split, the GA yields a 0.82 pp F1 improvement with minimal pruning (−4%), suggesting that the temporal distribution shift leaves little room for neuron removal. On CIC-IoT-2023 with the wider 96-bit thermometer and CE-leading weights, the GA instead expands the network: neurons grow from 153 to 221 (+44%) and bits per neuron from 43 to 64, an F1 gain of 1.92 pp (91.00→92.92%) and an FPR reduction of 7.72 pp (16.04→8.32%) that exploits the larger address space afforded by the 96-bit thermometer. Across all three datasets, the GA finds the per-dataset optimum directly rather than collapsing to a single architecture.

Since Vivado maps all sparse neuron memory to LUT-based distributed RAM (zero BRAM; Section 7), the neuron reduction translates directly to lower LUT utilization: the CICIDS GA genome uses 3,112 LUTs (5.9% of Z-7020) versus an estimated ∼5,500 for the unpruned grid-search genome.

Table 7. Phase progression across datasets (val_cal threshold, best-fitness genome). Neurons and bits are mean ± std.

Dataset	Phase	F1 (%)	FPR (%)	Neurons	Bits
CICIDS2017	Grid Search	99.19 ± 0.76	0.26 ± 0.31	343 ± 118	33 ± 5
	GA Neurons	99.30 ± 0.48	0.22 ± 0.12	198 ± 146	31 ± 5
	Δ	+0.11	−0.04	−145 (−42%)	
CIC-IoT-2023	Grid Search	91.00 ± 0.23	16.04 ± 1.44	153 ± 70	43 ± 14
	GA Neurons	92.92 ± 0.14	8.32 ± 0.70	221 ± 26	64 ± 3
	Δ	+1.92	−7.72	+68 (+44%)	+21
UNSW-NB15 (temporal)	Grid Search	86.52 ± 2.84	5.79 ± 4.81	348 ± 137	31 ± 1
	GA Neurons	87.34 ± 2.12	5.43 ± 4.02	333 ± 128	32 ± 2
	Δ	+0.82	−0.36	−15 (−4%)	

WNN model size is determined by the sparse memory footprint: only addresses visited during training store non-default cell values. Due to thermometer encoding correlation, our 32-bit address neurons contain only ∼3,500 trained entries each (Section 7), not the 2^{32} that a dense representation would require. Vivado maps these sparse tables entirely to LUT-based distributed RAM (zero BRAM), so the model is baked into the FPGA bitstream. Deployment sizes range from 2 LUTs (<0.01%, 5-neuron dense CIC-IoT) to 50,527 LUTs (95.0%, 500-neuron sparse CIC-IoT on the full 46M dataset), all fitting on a single $20 Zynq Z-7020.

7 Discussion

7.1 Connectivity as Feature Selection

During the grid search phase, each neuron's address bits are drawn randomly from the thermometer-encoded input vector. For example, 1,920 bits for CICIDS2017 (96-bit thermometer) or 152 bits for UNSW-NB15 (8-bit thermometer). Because this connectivity remains fixed throughout training, each neuron is permanently bound to a particular subset of input features. Then the genetic algorithm (GA) phase exploits this by optimizing neuron *count*: neurons whose random connectivity happens to capture discriminative input combinations survive, while those with uninformative connections are pruned. This results in an implicit form of feature selection driven entirely by neuron-level survival.

Analysis of our best CICIDS genome (F973, 11 neurons) reveals that each neuron's 34 address bits span 15–18 of the 20 input features, selecting 1–2 bits from each. This scattered connectivity contrasts with FWIW's random permutation, where each filter reads a contiguous slice of the input. Despite being randomly initialized, the surviving neurons collectively cover the most informative feature combinations, whereas the GA acts as a neuron-level selection mechanism rather than a connectivity optimizer.

The grid search phase establishes a strong baseline by exhaustively testing neuron-count and address-width combinations; the GA, seeded with the top-5 grid search configurations, then refines the architecture per dataset—pruning neuron count on the smaller splits and expanding it on the higher-volume CIC-IoT-2023—while maintaining or improving accuracy (Table 7).

7.2 Threshold Sensitivity

The gap between validation (val_cal) and deployable (train_cal) thresholds represents the central practical challenge for WNN-based IDS. On UNSW-NB15 temporal, this gap is 4.90 percentage points in F1 (88.87% vs. 83.97%) and 19.93 points in FPR (8.77% vs. 28.70%), driven by distribution shift between the time-ordered training partition (earlier traffic) and the test partition (later traffic), which moves the optimal decision threshold by an amount the training-set calibration cannot anticipate. Validation calibration samples the held-out distribution at deployment time to recover the target operating point. On CICIDS2017 random, the gap shrinks to 0.02 points (99.30% vs. 99.28%), confirming that random splits minimize calibration difficulty.

7.3 Encoding Resolution and Saturation

To determine the optimal thermometer encoding width, we swept from 2 to 64 bits per feature on all three datasets (2 runs per configuration). Figure 2 shows the results.

On CICIDS2017, an early sweep—predating the order-independent / QUAD-weighted pipeline fixes—had favored 16 bits. Re-run under the corrected pipeline and extended to 96 bits, the sweep instead selects **96-bit** as the strongest width: the 96-bit cohort reaches validation F1 99.55% (Table 4), improving on the earlier 16-bit operating point. The small residual gap to Random Forest (99.74%) narrows but persists across widths, confirming that precision alone does not explain the difference.

On UNSW-NB15 temporal, no clear inflection point emerges: F1 ranges from 82% to 90% across all encoding widths with 64-bit showing the most consistent results (89.56%). The high variance (only 2 runs per configuration) obscures any trend, and the harder temporal split amplifies seed sensitivity.

Address-tap saturation on CIC-IoT-2023. The best 32-bit and 34-bit peak genomes on the 46M dataset (245n×32b vs. 500n×34b) achieve F1 84.79% vs. 84.73%, a 0.06-point gap well within seed noise. The 4× larger address space yields no improvement, confirming that thermometer-encoding bit correlation caps effective discrimination: although each feature receives 8 thermometer bits, those bits are strongly correlated — only 9 distinct patterns (0 through 8 consecutive ones) are ever observed — so each feature contributes only $\log_2(9) \approx 3.17$ bits of entropy. Across 20 features that is roughly 63 bits of discriminative information, of which a 32-bit address tap already samples about half; adding more tap bits merely re-samples already-represented information.

To test whether this saturation is bounded by the 8-bit thermometer width, we re-trained both peak architectures at 16/32/64-bit thermometer encodings on 46M (2 seeds per cell, Table 8). Neither architecture shows meaningful F1 movement: the entire sweep lands within ±0.5 pp of the 8-bit baseline, well inside seed noise. This rules out both interpretations of the saturation: neither wider address taps nor higher-resolution encoding unlocks additional F1.

Table 8. 46M thermometer encoding sweep on two peak architectures (train_cal threshold, best-fitness genome). The 8-bit columns are the baselines from Table 6; 16/32/64-bit cells report mean ± std across 2 seeds.

Architecture	Thermometer width			
	8-bit	16-bit	32-bit	64-bit
F1 (%)				
96n × 32b	84.36	84.13 ± 0.08	84.53 ± 0.25	84.59 ± 0.13
500n × 34b	84.73	84.43 ± 0.21	84.46 ± 0.41	84.87 ± 0.03
FPR (%)				
96n × 32b	1.61	1.17 ± 0.22	1.50 ± 0.27	1.69 ± 0.17
500n × 34b	1.91	1.51 ± 0.05	1.37 ± 0.18	1.75 ± 0.02
Accuracy (%)				
96n × 32b	97.59	97.53 ± 0.02	97.63 ± 0.06	97.64 ± 0.03
500n × 34b	97.68	97.60 ± 0.05	97.61 ± 0.09	97.70 ± 0.01

Scale-dependent 2-bit finding. A downward sweep (2/4-bit) on the same architectures revealed an unexpected lift at 2-bit on 46M (+1.26 pp at 96n × 32b, +1.04 pp at 400n × 8b). However, a 4-seed validation at the 1.3M scale showed a balanced-fitness drop of 0.54 pp, outside the ±0.3 pp noise floor. The lift appears to be training-set-size-dependent: narrower encodings benefit from the examples-per-address density that 46M provides but 1.3M does not.

7.4 FPGA Deployment

RAM neurons map naturally to FPGA resources: each neuron's lookup table is synthesized into distributed LUTs with the evolved connection map implemented as compile-time wire assignments (pure routing, zero logic). We synthesize all

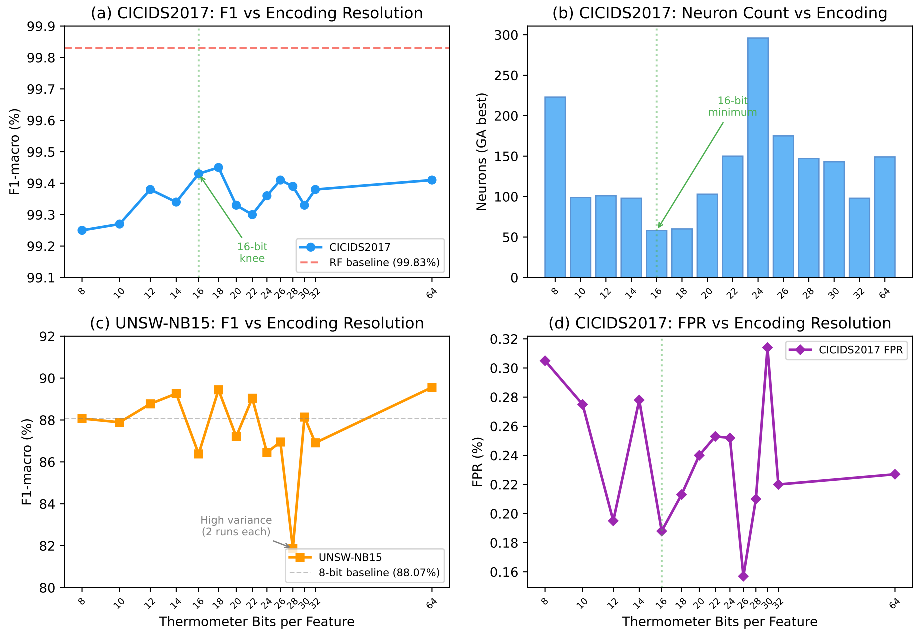

Fig. 2. Effect of thermometer encoding resolution on classification performance. (a) CICIDS2017 F1 shows a clear knee at 16 bits, with diminishing returns beyond. (b) The GA finds the fewest neurons at 16 bits, suggesting optimal information density. (c) UNSW-NB15 temporal split shows no clear trend (high variance from only 2 runs per configuration). (d) CICIDS2017 false positive rate is lowest around 16–26 bits.

best genomes from Tables 2–6 on the Xilinx Zynq Z-7020 (53,200 LUTs, 140 BRAM36 blocks) and report LUT utilization, latency, and power in each result table.

A key finding is that **Vivado infers zero BRAMs for all configurations**: both dense and sparse neuron memories are mapped entirely to LUT-based distributed RAM, with the trained model baked into the FPGA bitstream. This enables deployment of even our largest genome (500n $\times$ 34b, 50,527 LUTs, 95% of the Z-7020) on a single chip with no external memory.

Dense vs. sparse deployment. Our architecture supports two deployment modes. *Dense* ($O(1)$) mode caps address width at 8–12 bits, storing each neuron's full lookup table in distributed LUTs for single-cycle inference. On CIC-IoT-2023, the 5n$\times$4b genome requires just 2 LUTs with 2 ns latency, while the 400n$\times$8b peak genome uses 304 LUTs at 15 ns (Table 6). *Sparse* ($O(\log n)$) mode uses the full 32–34-bit address space. Thermometer encoding creates massive address

The XC7Z020-CLG400 is available on surplus EBAZ4206 boards (decommissioned Bitcoin miner control boards) for under \$25; purpose-built development boards (Arty Z7, ALINX AX7020) range from \$200–350. Our synthesis targets the chip, not a specific board; utilization numbers are identical on any Z-7020 platform.

collisions: each bit within a feature is correlated (bit k is set iff the value exceeds threshold k), so a 32-bit neuron observes only ~3,500 unique addresses from millions of training examples, not 2^{32}. Vivado maps these sorted arrays to distributed LUTs, achieving 52–227 ns latency across all datasets (Tables 2–5).

Comparison with FWIW. FWIW [21] achieves a compact footprint (269 LUTs on Virtex UltraScale+ at 740 MS/s throughput) through Bloom filter neurons with random connections on a 90/10 deduplicated UNSW-NB15 split with 1:1 attack oversampling in training. Our dense 47-neuron, 12-bit genome uses just **33 LUTs** (8.2× fewer) at 9.9 ns latency on the Zynq Z-7020, with 93.97% F1, 0.62% FPR, and 99.09% accuracy — exceeding FWIW's reported 98.5% accuracy (unbalanced test set) and its 1.57% FPR (reported only on the 2:1 balanced split, at 98.9% accuracy), albeit on different partitions of the same dataset. Our sparse genomes push further to 94.86% F1, 0.52% FPR, 99.23% accuracy (Table 3).

Latency and throughput. The dense (≤8-bit) genomes achieve true sub-microsecond on-chip latency: 2–30 ns for an $O(1)$ direct lookup, fully resident in the Z-7020's distributed RAM (Tables 2–6). The sparse (≥32-bit) genomes use an $O(\log n)$ binary search at **3 FSM cycles per probe** (verified cycle-accurately in Vivado xsim: 49–58 cycles per classification); at the synthesized clock this is ~0.6–0.85 μs when the sorted tables fit on-chip. Those tables, however, range 3.4–18 MB and exceed the Z-7020's 0.63 MB BRAM, so on a Z-7020 they are external-DRAM-backed: with all N neurons' dependent probes serialized through one DDR port, a cycle-accurate arbiter model gives a contention-bound latency of ~20–90 μs. The dense tier is thus the genuinely fully-on-chip, sub-microsecond design point; the sparse high-accuracy tier trades that for either a larger (UltraScale+) device or DRAM-backed deployment. Beyond per-packet latency, another relevant metric for network IDS is throughput: 100 Gbps line-rate inspection requires only ~12.5 MS/s (at ~1,000-byte average packets). The **dense** tier meets this comfortably—its 2–30 ns $O(1)$ lookup sustains ~33–500 MS/s. The sparse classifiers process one query at a time (the binary-search FSM is not pipelined), so their throughput is latency-bound: ~1.6 MS/s on-chip and far less when DRAM-backed—below 100 Gbps line rate. A pipelined search or the hash-based lookup discussed in Section 7.5 would lift sparse throughput toward the dense tier's. For the dense tier, the sub-microsecond classification latency is negligible compared to feature-extraction overhead, which typically dominates the packet-processing pipeline.

Functional verification. We verified both sparse and dense RTL implementations against the Python training model using Vivado xsim 2025.2. Testbenches feed 1,000 stratified test vectors (500 normal, 500 attack) from both CICIDS2017 and UNSW-NB15 test sets through the synthesized classifiers and compare output scores against Python-computed expected values. Both configurations produce **identical scores for all 1,000 vectors** (100% match on each dataset), confirming that the FPGA implementations faithfully reproduce the trained QSR models.

7.5 Future Directions

Several directions offer potential for further improvement. First, extending to *multi-class classification* (one discriminator per attack category) would exploit the rich multi-class labels available in UNSW-NB15, CICIDS2017, and CIC-IoT-2023 (10, 14, and 33 attack categories respectively), enabling attack-type identification beyond binary detection.

Second, our dense FPGA configurations achieve 2–30 ns $O(1)$ latency (Table 6), while sparse configurations need $\sim$0.6–0.85 μs on-chip ($O(\log n)$ binary search at 3 FSM cycles per probe; Tables 2–5) and, when DRAM-backed, $\sim$20–90 μs under single-port contention. A *hash-based lookup* replacing binary search for sparse genomes could reduce both the on-chip latency to near-$O(1)$ *and* the DRAM probe count (one lookup instead of $\log n$ dependent reads), combining the accuracy of large address spaces with dense-like speed and slashing the DRAM-contention cost.

Third, our current framework applies a uniform thermometer width to all features. However, feature cardinality varies widely across IDS datasets: in UNSW-NB15, `ct_state_ttl` has only 7 unique values (needing just 6 bits for lossless encoding) while `Sjit` has over 857K (heavily quantized at any practical width). *Per-feature adaptive thermometer encoding* would allocate bits proportional to each feature's cardinality, reducing wasted resolution on low-cardinality features while improving fidelity on continuous ones.

Fourth, the remaining 0.31% F1 gap to Random Forest on CICIDS2017 (0.19% for the best genome) is not explained by encoding precision (64-bit yields no improvement) and warrants further investigation. Our framework supports additional optimization phases beyond the two evaluated here: *GA bits* (evolving address width per neuron), *GA connections* (evolving which input bits each neuron reads), *tabu search* variants of each (TS neurons, TS bits, TS connections), and *Lamarckian genesis phases* (neurogenesis, synaptogenesis, axonogenesis) that progressively refine neuron structure, all already implemented in our framework. These phases, combined with multi-cluster discriminators and hybrid encoding strategies, offer multiple paths to close the accuracy gap while preserving the hardware-friendly lookup-based architecture.

Finally, we observed during the PUB50 batch that best-FPR genomes occasionally converge on radically smaller architectures than the best-fitness population (e.g., 6 neurons $\times$ 16-bit address producing 1.09% FPR on CIC-IoT-2023 under train_cal), at the cost of lower F1. These extreme-compression outliers suggest two complementary *multi-genome ensemble* strategies. (1) A *sequential cascade*: an ultra-small pre-filter WNN eliminates the majority of benign traffic at near-zero FPR, forwarding only uncertain or attack-likely flows to a larger second-stage classifier with higher F1. (2) A *parallel mixture-of-experts*: multiple heterogeneous genomes (low-FPR, high-F1, high-accuracy variants) run concurrently and combine their outputs via weighted voting or learned gating. Because single-neuron lookup is $O(1)$ and sub-microsecond, both designs remain within the line-rate budget even with two or three genomes active simultaneously. A well-chosen pairing could achieve strictly better F1/FPR/Acc than any single genome in the population, amortizing per-packet cost across the deployment while preserving the high-F1 operating point for the flows that need it.

8 Conclusion

We presented an evolutionary weightless neural network architecture for network intrusion detection, combining Quad-State RAM neurons with genetic algorithm optimization of neuron connectivity. Our approach contributes three advances over prior WNN-based IDS:

1. **Evolved connectivity.** A two-phase pipeline (grid search + GA neuron optimization) discovers compact, accurate genomes. The GA reduces neuron count by 42% while improving F1 and halving variance compared to grid search alone.
2. **Three-dataset evaluation with ML baselines.** We evaluate on UNSW-NB15, CICIDS2017, and CIC-IoT-2023, comparing directly against Random Forest and XGBoost on identical splits. On UNSW-NB15 (temporal, 30 runs, 16-bit thermometer), our WNN outperforms both baselines in F1 (88.87% vs. 86.05%) at roughly one-third the FPR (8.77% vs. 25.41%), with the best single seed reaching 89.34% F1 at 9.76% FPR. A 5-neuron Pareto operating point from the same evolutionary population reaches 87.88% F1 at 6.94% FPR in approximately 2 KB of model memory, fitting the Zynq Z-7020 internal BRAM with 99% headroom. On CICIDS2017 (random, 112 runs), our best genome reaches 99.55% F1, within 0.19 pp of RF. On CIC-IoT-2023 (random, 30 runs cohort complete as of 03/07/2026, canonical TOP20, 96-bit thermometer, CE-heavy weights), the WNN's GA Neurons mean F1 (92.81%) exceeds AdaBoost (top-20) at 91.61% while simultaneously cutting FPR (8.57% vs. 11.77%) and matches XGBoost (93.36%) on F1 within 0.55 pp at FPR lower by 3.08 pp. At the fixed-0.5 threshold, the GA Neurons cohort drives FPR below 1% (0.96±0.11%) at mean F1 of 86.43±0.88% — a sub-1% mean-FPR operating point unreachable by tree ensembles at any threshold (Random Forest's lowest FPR with F1>80% is 9.77%, XGBoost's is 11.65%), a critical advantage for operational deployment where false-positive volume determines alert-handling cost.
3. **FPGA deployment.** The dense $O(1)$ lookup configurations deploy fully on-chip on a $20 Zynq Z-7020 with genuine sub-microsecond latency (2–30 ns) that sustains 100 Gbps line-rate throughput, using as few as 2 LUTs (<0.01%); the CIC-IoT-2023 peak genome (400n × 8b, 83.13% F1) fits in 304 LUTs at 67 MHz in 25 KB of model storage. The wider sparse-search genomes (32–64 bit) exceed the Z-7020's 0.63 MB on-chip BRAM: their tables must be DRAM-backed, where single-port contention over the dependent binary-search probes raises latency to ~20–90 μs — still well within edge-IDS budgets, but no longer line-rate.

Future work includes multi-class attack categorization, UNSW-NB15 random-split evaluation with balanced fitness weights, adaptive threshold calibration, and scaling the evolutionary search to the full 46M CIC-IoT-2023 dataset to close the gap with tree ensembles.

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A Full Threshold Mode Breakdown

This appendix reports the full seven-mode threshold calibration breakdown for every dataset across all five genome-selection criteria: *best-fitness* (by Equation 2), *best-F1*, *best-FPR*, *best-Acc*, and *best-CE*. For each dataset–genome combination, we report both Grid Search and GA Neurons phases as mean ± std across all completed flows. The main body (Tables 2, 3, 4, 5) summarises only three modes (train-cal, fixed 0.5, val-cal) for the best-fitness GA genome; the tables below show how those trends hold across the other six criteria and the full seven-mode sweep.

Table 9. UNSW-NB15 temporal, best-fitness genome (30 runs, 16-bit thermometer). Grid Search: 114±162 neurons / 33±2 bits; GA Neurons: 131±158 neurons / 33±2 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	83.22±1.25	83.92±1.10	30.36±3.53	28.52±3.14	84.01±1.03	84.60±0.93
Fixed 0.5	84.39±1.88	84.34±1.67	26.25±5.61	26.63±4.78	84.98±1.57	84.94±1.40
Platt	83.87±1.56	84.28±1.26	28.03±4.40	27.25±3.47	84.54±1.30	84.90±1.08
Beta	79.60±3.53	79.67±3.03	39.17±8.34	39.21±7.20	81.15±2.80	81.19±2.40
Empirical	83.34±2.78	83.24±2.23	26.85±12.30	27.84±11.14	84.10±2.30	84.02±1.79
Empirical cumul.	85.79±1.95	85.57±1.95	20.60±6.18	20.90±6.28	86.14±1.72	85.93±1.73
Val_cal	86.47±1.95	85.88±2.02	14.62±6.54	17.05±7.35	86.64±1.83	86.13±1.82

Table 10. UNSW-NB15 temporal, best-F1 genome (30 runs, 16-bit thermometer). Grid Search: 114±162 neurons / 33±2 bits; GA Neurons: 131±158 neurons / 33±2 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	83.22±1.25	83.92±1.10	30.36±3.53	28.52±3.14	84.01±1.03	84.60±0.93
Fixed 0.5	84.39±1.88	84.34±1.67	26.25±5.61	26.63±4.78	84.98±1.57	84.94±1.40
Platt	83.87±1.56	84.28±1.26	28.03±4.40	27.25±3.47	84.54±1.30	84.90±1.08
Beta	79.60±3.53	79.67±3.03	39.17±8.34	39.21±7.20	81.15±2.80	81.19±2.40
Empirical	83.34±2.78	83.24±2.23	26.85±12.30	27.84±11.14	84.10±2.30	84.02±1.79
Empirical cumul.	85.79±1.95	85.57±1.95	20.60±6.18	20.90±6.28	86.14±1.72	85.93±1.73
Val_cal	86.47±1.95	85.88±2.02	14.62±6.54	17.05±7.35	86.64±1.83	86.13±1.82

Table 11. UNSW-NB15 temporal, best-FPR genome (30 runs, 16-bit thermometer). Grid Search: 238±167 neurons / 21±13 bits; GA Neurons: 165±173 neurons / 22±13 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	79.78±5.68	79.64±5.49	35.38±11.39	35.86±11.22	81.07±4.77	80.96±4.64
Fixed 0.5	81.24±5.28	80.75±4.41	23.95±4.06	25.99±6.06	81.61±5.49	81.24±4.63
Platt	80.41±5.76	79.74±5.71	31.81±8.77	33.99±8.27	81.27±5.33	80.74±5.27
Beta	77.67±7.03	77.45±6.06	40.20±11.49	41.77±10.43	79.36±5.83	79.29±4.87
Empirical	80.79±7.24	78.97±8.04	24.40±22.31	33.20±22.56	81.92±5.79	80.68±6.33
Empirical cumul.	84.14±5.67	83.38±5.77	9.34±3.37	9.22±6.80	84.26±5.62	83.59±5.58
Val_cal	84.47±5.46	84.28±4.97	7.96±2.49	6.75±3.37	84.55±5.41	84.38±4.90

Table 12. UNSW-NB15 temporal, best-Acc genome (30 runs, 16-bit thermometer). Grid Search: 120±152 neurons / 33±2 bits; GA Neurons: 131±158 neurons / 33±2 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	82.84±1.29	83.99±1.43	31.19±3.45	28.33±4.03	83.68±1.08	84.67±1.21
Fixed 0.5	83.55±1.84	84.30±1.66	28.58±5.30	26.77±4.76	84.25±1.55	84.90±1.41
Platt	83.19±1.45	84.13±1.30	29.82±3.88	27.72±3.53	83.94±1.22	84.77±1.11
Beta	78.88±3.39	79.60±3.08	40.92±7.55	39.34±7.24	80.58±2.68	81.14±2.44
Empirical	82.85±2.87	83.05±2.38	28.96±11.38	28.29±11.44	83.70±2.37	83.86±1.91
Empirical cumul.	84.90±2.21	85.48±1.98	23.19±6.96	20.62±6.37	85.36±1.94	85.83±1.78
Val_cal	85.37±2.25	85.80±2.04	18.37±8.10	16.80±7.32	85.66±2.05	86.04±1.84

Table 13. UNSW-NB15 temporal, best-CE genome (30 runs, 16-bit thermometer). Grid Search: 410±99 neurons / 34±1 bits; GA Neurons: 309±107 neurons / 33±1 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	83.90±0.24	83.97±0.34	28.76±0.75	28.71±1.04	84.58±0.20	84.65±0.29
Fixed 0.5	84.95±0.19	84.27±0.44	25.45±0.54	27.74±1.25	85.46±0.17	84.90±0.37
Platt	84.31±0.14	84.12±0.24	27.45±0.34	28.23±0.59	84.92±0.12	84.78±0.21
Beta	83.29±0.11	83.26±0.21	30.64±0.25	30.92±0.48	84.08±0.10	84.06±0.18
Empirical	85.74±1.62	87.18±1.14	5.74±7.08	5.57±4.51	85.80±1.36	87.22±1.11
Empirical cumul.	88.03±0.70	88.45±0.34	12.99±2.74	12.40±1.04	88.15±0.62	88.56±0.33
Val_cal	88.54±0.10	88.86±0.24	9.12±0.58	8.78±1.18	88.59±0.10	88.91±0.24

Table 14. UNSW-NB15 random, best-fitness genome (112 runs, 16-bit thermometer). Grid Search: 370±123 neurons / 33±2 bits; GA Neurons: 346±136 neurons / 32±2 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	94.36±0.14	94.39±0.15	0.44±0.05	0.45±0.05	99.17±0.02	99.17±0.02
Fixed 0.5	93.78±0.09	93.76±0.11	1.02±0.03	1.03±0.04	98.98±0.02	98.98±0.02
Platt	94.40±0.14	94.44±0.13	0.54±0.07	0.53±0.08	99.16±0.02	99.16±0.03
Beta	94.38±0.24	94.39±0.21	0.57±0.09	0.56±0.10	99.15±0.04	99.15±0.03
Empirical	93.39±0.18	93.32±0.22	1.14±0.04	1.16±0.05	98.90±0.04	98.88±0.04
Empirical cumul.	94.35±0.22	94.38±0.17	0.51±0.10	0.51±0.12	99.16±0.03	99.16±0.03
Val_cal	94.46±0.12	94.49±0.14	0.48±0.06	0.49±0.05	99.18±0.02	99.18±0.02

Table 15. UNSW-NB15 random, best-F1 genome (112 runs, 16-bit thermometer). Grid Search: 380±118 neurons / 33±2 bits; GA Neurons: 345±136 neurons / 32±2 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	94.36±0.15	94.38±0.16	0.44±0.05	0.44±0.05	99.17±0.02	99.17±0.02
Fixed 0.5	93.78±0.09	93.75±0.12	1.02±0.03	1.03±0.04	98.98±0.02	98.98±0.03
Platt	94.39±0.19	94.42±0.14	0.54±0.08	0.53±0.08	99.16±0.03	99.16±0.03
Beta	94.39±0.21	94.38±0.24	0.56±0.09	0.55±0.10	99.15±0.03	99.15±0.04
Empirical	93.37±0.19	93.33±0.22	1.15±0.04	1.15±0.05	98.89±0.04	98.89±0.04
Empirical cumul.	94.34±0.24	94.38±0.16	0.51±0.10	0.51±0.12	99.16±0.03	99.16±0.03
Val_cal	94.46±0.12	94.49±0.13	0.48±0.06	0.48±0.05	99.18±0.02	99.18±0.02

Table 16. UNSW-NB15 random, best-FPR genome (112 runs, 16-bit thermometer). Grid Search: 359±123 neurons / 31±3 bits; GA Neurons: 345±148 neurons / 32±1 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	94.32±0.18	94.43±0.12	0.45±0.06	0.45±0.05	99.16±0.03	99.18±0.02
Fixed 0.5	93.75±0.10	93.77±0.12	1.03±0.04	1.02±0.04	98.97±0.02	98.98±0.03
Platt	94.33±0.25	94.40±0.17	0.54±0.09	0.53±0.08	99.15±0.04	99.16±0.03
Beta	94.32±0.28	94.41±0.18	0.56±0.08	0.55±0.09	99.14±0.04	99.16±0.03
Empirical	93.35±0.21	93.32±0.21	1.15±0.04	1.16±0.05	98.89±0.04	98.88±0.04
Empirical cumul.	94.26±0.33	94.34±0.24	0.51±0.10	0.51±0.11	99.15±0.04	99.16±0.04
Val_cal	94.36±0.14	94.47±0.13	0.50±0.07	0.48±0.05	99.16±0.02	99.18±0.02

Table 17. UNSW-NB15 random, best-Acc genome (112 runs, 16-bit thermometer). Grid Search: 366±119 neurons / 33±2 bits; GA Neurons: 350±136 neurons / 32±2 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	94.35±0.13	94.40±0.14	0.44±0.05	0.44±0.04	99.17±0.02	99.18±0.02
Fixed 0.5	93.77±0.09	93.77±0.10	1.02±0.03	1.02±0.04	98.98±0.02	98.98±0.02
Platt	94.40±0.18	94.44±0.13	0.53±0.08	0.53±0.09	99.16±0.03	99.17±0.03
Beta	94.39±0.24	94.41±0.14	0.57±0.08	0.56±0.09	99.15±0.03	99.16±0.03
Empirical	93.41±0.17	93.38±0.19	1.14±0.04	1.15±0.04	98.90±0.03	98.89±0.04
Empirical cumul.	94.34±0.23	94.34±0.28	0.51±0.11	0.52±0.12	99.16±0.03	99.16±0.04
Val_cal	94.46±0.13	94.47±0.13	0.48±0.06	0.48±0.05	99.18±0.02	99.18±0.02

Table 18. UNSW-NB15 random, best-CE genome (112 runs, 16-bit thermometer). Grid Search: 369±127 neurons / 33±1 bits; GA Neurons: 333±147 neurons / 32±2 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	94.37±0.12	94.43±0.14	0.45±0.04	0.45±0.05	99.17±0.02	99.18±0.02
Fixed 0.5	93.78±0.10	93.77±0.13	1.02±0.03	1.02±0.04	98.98±0.02	98.98±0.03
Platt	94.42±0.17	94.40±0.16	0.54±0.07	0.52±0.09	99.16±0.03	99.16±0.03
Beta	94.42±0.13	94.42±0.14	0.58±0.08	0.54±0.09	99.15±0.03	99.16±0.03
Empirical	93.36±0.18	93.36±0.20	1.15±0.04	1.15±0.04	98.89±0.04	98.89±0.04
Empirical cumul.	94.36±0.21	94.38±0.22	0.53±0.09	0.50±0.09	99.16±0.03	99.16±0.03
Val_cal	94.44±0.13	94.48±0.14	0.48±0.05	0.47±0.05	99.17±0.02	99.18±0.02

Table 19. CICIDS2017 random, best-fitness genome (112 runs, 16-bit thermometer). Grid Search: 346±113 neurons / 34±1 bits; GA Neurons: 200±148 neurons / 32±1 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	99.34±0.09	99.38±0.10	0.20±0.07	0.20±0.08	99.58±0.06	99.61±0.06
Fixed 0.5	99.25±0.12	99.22±0.14	0.35±0.07	0.41±0.11	99.53±0.08	99.51±0.09
Platt	99.22±0.18	99.29±0.13	0.30±0.12	0.30±0.10	99.51±0.11	99.55±0.08
Beta	99.25±0.17	99.29±0.17	0.28±0.10	0.27±0.10	99.53±0.11	99.55±0.11
Empirical	98.24±0.70	98.64±0.59	1.29±0.62	0.95±0.52	98.87±0.47	99.13±0.39
Empirical cumul.	99.21±0.23	99.31±0.12	0.26±0.15	0.24±0.09	99.50±0.14	99.57±0.08
Val_cal	99.35±0.09	99.40±0.06	0.21±0.07	0.20±0.06	99.59±0.05	99.62±0.04

Table 20. CICIDS2017 random, best-F1 genome (112 runs, 16-bit thermometer). Grid Search: 337±117 neurons / 34±1 bits; GA Neurons: 197±146 neurons / 32±1 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	99.33±0.11	99.38±0.10	0.20±0.07	0.20±0.08	99.58±0.07	99.61±0.06
Fixed 0.5	99.25±0.14	99.22±0.13	0.35±0.09	0.41±0.11	99.53±0.09	99.51±0.09
Platt	99.23±0.17	99.29±0.13	0.30±0.12	0.30±0.11	99.52±0.10	99.55±0.08
Beta	99.24±0.19	99.30±0.15	0.28±0.11	0.27±0.10	99.52±0.12	99.56±0.10
Empirical	98.29±0.72	98.65±0.58	1.25±0.63	0.94±0.52	98.89±0.48	99.13±0.39
Empirical cumul.	99.22±0.22	99.31±0.12	0.25±0.14	0.24±0.09	99.51±0.13	99.57±0.08
Val_cal	99.34±0.09	99.40±0.06	0.22±0.08	0.20±0.06	99.59±0.06	99.62±0.04

Table 21. CICIDS2017 random, best-FPR genome (112 runs, 16-bit thermometer). Grid Search: 346±123 neurons / 33±1 bits; GA Neurons: 228±167 neurons / 32±1 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	99.30±0.14	99.31±0.13	0.22±0.09	0.23±0.09	99.56±0.09	99.57±0.08
Fixed 0.5	99.23±0.12	99.18±0.22	0.35±0.07	0.41±0.14	99.51±0.07	99.48±0.14
Platt	99.23±0.16	99.24±0.20	0.30±0.08	0.31±0.15	99.51±0.10	99.52±0.13
Beta	99.24±0.17	99.23±0.20	0.28±0.10	0.28±0.09	99.52±0.11	99.52±0.12
Empirical	98.20±0.72	98.47±0.84	1.32±0.63	1.07±0.75	98.84±0.48	99.02±0.57
Empirical cumul.	99.21±0.19	99.18±0.32	0.25±0.12	0.28±0.16	99.50±0.12	99.48±0.20
Val_cal	99.33±0.08	99.31±0.17	0.22±0.07	0.23±0.10	99.57±0.05	99.56±0.11

Table 22. CICIDS2017 random, best-Acc genome (112 runs, 16-bit thermometer). Grid Search: 338±116 neurons / 34±1 bits; GA Neurons: 199±148 neurons / 32±1 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	99.33±0.11	99.38±0.10	0.20±0.07	0.20±0.08	99.58±0.07	99.61±0.06
Fixed 0.5	99.25±0.14	99.22±0.13	0.35±0.09	0.41±0.11	99.53±0.09	99.51±0.09
Platt	99.23±0.17	99.29±0.13	0.30±0.12	0.30±0.11	99.52±0.10	99.55±0.08
Beta	99.24±0.19	99.30±0.15	0.28±0.10	0.27±0.09	99.52±0.12	99.56±0.10
Empirical	98.29±0.72	98.66±0.58	1.25±0.63	0.93±0.51	98.89±0.48	99.14±0.38
Empirical cumul.	99.22±0.21	99.31±0.12	0.25±0.14	0.24±0.09	99.51±0.13	99.57±0.08
Val_cal	99.34±0.10	99.40±0.06	0.22±0.08	0.20±0.06	99.58±0.06	99.62±0.04

Table 23. CICIDS2017 random, best-CE genome (112 runs, 16-bit thermometer). Grid Search: 276±142 neurons / 34±1 bits; GA Neurons: 181±139 neurons / 32±1 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	99.28±0.19	99.39±0.08	0.24±0.13	0.21±0.09	99.55±0.12	99.61±0.05
Fixed 0.5	99.14±0.38	99.17±0.15	0.44±0.28	0.48±0.13	99.45±0.24	99.47±0.09
Platt	99.19±0.25	99.29±0.11	0.32±0.16	0.32±0.09	99.48±0.16	99.55±0.07
Beta	99.17±0.32	99.31±0.11	0.32±0.22	0.27±0.08	99.48±0.21	99.56±0.07
Empirical	98.30±0.66	98.63±0.56	1.21±0.62	0.95±0.51	98.90±0.44	99.12±0.37
Empirical cumul.	99.14±0.30	99.29±0.16	0.32±0.24	0.24±0.10	99.46±0.19	99.55±0.10
Val_cal	99.30±0.17	99.38±0.09	0.23±0.10	0.21±0.07	99.56±0.11	99.61±0.06

Table 24. CIC-IoT-2023 random (1.14M), best-fitness genome (96-bit thermometer, CE-heavy weights, 250n×100b architecture ceiling, canonical TOP20; 30 runs (cohort complete) as of 03/07/2026). Grid Search: 233±24 neurons / 64±0 bits; GA Neurons: 228±18 neurons / 64±2 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	90.16±0.16	92.81±0.17	15.29±0.67	8.57±0.67	95.19±0.11	96.42±0.08
Fixed 0.5	81.96±0.10	86.43±0.88	1.12±0.04	0.96±0.11	88.67±0.08	92.04±0.62
Platt	90.11±0.13	92.80±0.17	12.71±0.20	8.76±0.46	95.04±0.07	96.42±0.09
Beta	90.14±0.17	92.67±0.16	16.01±0.23	10.68±0.56	95.21±0.09	96.41±0.08
Empirical	89.84±0.24	92.63±0.19	20.17±0.67	11.04±0.79	95.24±0.11	96.40±0.08
Empirical cumul.	89.78±0.15	92.56±0.21	10.17±0.32	5.57±0.58	94.73±0.10	96.18±0.12
Val_cal	90.18±0.16	92.83±0.17	14.72±0.84	8.27±0.58	95.18±0.11	96.42±0.09

Table 25. CIC-IoT-2023 random (1.14M), best-F1 genome (96-bit thermometer, CE-heavy weights, 250n×100b architecture ceiling, canonical TOP20; 30 runs (cohort complete) as of 03/07/2026). Grid Search: 98±66 neurons / 42±17 bits; GA Neurons: 200±60 neurons / 60±6 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	90.94±0.25	92.92±0.15	15.50±1.36	8.63±0.65	95.64±0.11	96.48±0.07
Fixed 0.5	80.63±0.70	85.01±1.64	1.05±0.15	0.87±0.11	87.57±0.59	90.98±1.22
Platt	89.88±0.64	92.90±0.15	11.27±0.40	8.90±0.62	94.83±0.38	96.48±0.07
Beta	90.43±1.02	92.76±0.18	18.23±4.76	10.84±0.93	95.49±0.31	96.46±0.08
Empirical	90.82±0.25	92.79±0.15	17.21±1.26	10.56±0.77	95.64±0.11	96.47±0.07
Empirical cumul.	90.75±0.36	92.74±0.15	13.68±1.86	6.35±1.18	95.45±0.22	96.31±0.09
Val_cal	90.95±0.25	92.93±0.15	15.33±1.39	8.34±0.84	95.63±0.11	96.48±0.07

Table 26. CIC-IoT-2023 random (1.14M), best-FPR genome (96-bit thermometer, CE-heavy weights, 250n×100b architecture ceiling, canonical TOP20; 30 runs (cohort complete) as of 03/07/2026). Grid Search: 124±77 neurons / 67±9 bits; GA Neurons: 222±28 neurons / 64±3 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	89.60±0.97	92.83±0.18	14.61±0.96	8.51±0.64	94.83±0.55	96.43±0.09
Fixed 0.5	82.00±0.30	86.31±0.94	1.25±0.46	0.92±0.10	88.72±0.25	91.95±0.67
Platt	89.61±0.97	92.82±0.17	13.08±1.22	8.68±0.50	94.76±0.52	96.43±0.09
Beta	89.29±1.76	92.69±0.18	17.22±4.38	10.58±0.61	94.80±0.75	96.42±0.09
Empirical	89.27±0.92	92.66±0.19	19.13±1.31	10.89±0.76	94.86±0.56	96.41±0.09
Empirical cumul.	89.16±1.25	92.56±0.20	9.34±1.05	5.37±0.54	94.29±0.87	96.17±0.11
Val_cal	89.63±0.95	92.84±0.17	13.63±0.98	8.05±0.81	94.80±0.55	96.42±0.09

Table 27. CIC-IoT-2023 random (1.14M), best-Acc genome (96-bit thermometer, CE-heavy weights, 250n×100b architecture ceiling, canonical TOP20; 30 runs (cohort complete) as of 03/07/2026). Grid Search: 103±72 neurons / 42±17 bits; GA Neurons: 195±63 neurons / 59±8 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	90.91±0.28	92.89±0.18	15.54±1.41	8.89±0.89	95.62±0.12	96.47±0.08
Fixed 0.5	80.64±0.70	84.63±1.80	1.06±0.14	0.87±0.09	87.58±0.58	90.70±1.37
Platt	89.81±0.67	92.85±0.27	11.31±0.40	8.97±0.61	94.80±0.40	96.45±0.14
Beta	90.51±0.71	92.73±0.20	17.89±3.77	11.06±1.05	95.51±0.22	96.45±0.08
Empirical	90.81±0.27	92.77±0.17	17.21±1.26	10.67±0.74	95.64±0.12	96.47±0.07
Empirical cumul.	90.75±0.36	92.73±0.16	14.01±2.00	6.84±1.59	95.46±0.22	96.32±0.08
Val_cal	90.92±0.28	92.90±0.18	15.43±1.45	8.68±1.03	95.62±0.12	96.47±0.08

Table 28. CIC-IoT-2023 random (1.14M), best-CE genome (96-bit thermometer, CE-heavy weights, 250n×100b architecture ceiling, canonical TOP20; 30 runs (cohort complete) as of 03/07/2026). Identical to the best-fitness genome: under CE-heavy weights the fitness ranking is CE-dominated. Grid Search: 233±24 neurons / 64±0 bits; GA Neurons: 228±18 neurons / 64±2 bits.

Threshold	F1 Grid (%)	F1 GA (%)	FPR Grid (%)	FPR GA (%)	Acc Grid (%)	Acc GA (%)
Train-cal	90.16±0.16	92.81±0.17	15.29±0.67	8.57±0.67	95.19±0.11	96.42±0.08
Fixed 0.5	81.96±0.10	86.43±0.88	1.12±0.04	0.96±0.11	88.67±0.08	92.04±0.62
Platt	90.11±0.13	92.80±0.17	12.71±0.20	8.76±0.46	95.04±0.07	96.42±0.09
Beta	90.14±0.17	92.67±0.16	16.01±0.23	10.68±0.56	95.21±0.09	96.41±0.08
Empirical	89.84±0.24	92.63±0.19	20.17±0.67	11.04±0.79	95.24±0.11	96.40±0.08
Empirical cumul.	89.78±0.15	92.56±0.21	10.17±0.32	5.57±0.58	94.73±0.10	96.18±0.12
Val_cal	90.18±0.16	92.83±0.17	14.72±0.84	8.27±0.58	95.18±0.11	96.42±0.09